

Quantum Poisson Surface Reconstruction

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1 Introduction

1.1 Background and Motivation

Poisson Surface Reconstruction (PSR) is an important method in computer science that generates a continuous surface from a point cloud and its corresponding surface normals. The method has widespread applications in areas such as 3D scanning, computer vision, and computer graphics, including the creation and processing of 3D models used in animated films such as Toy Story and Shrek. However, one of the main challenges of PSR is its computational and memory requirements. 3D scanners can produce point clouds that contain billions of points. Furthermore, achieving a high-resolution reconstruction requires a finer grid, which increases the number of computations and memory required. As a result, reconstructing high-resolution surfaces from large point clouds can become computationally expensive and memory intensive, highlighting the need for more efficient PSR methods.

Quantum Computing is an emerging field that offers the potential to solve certain computational problems more efficiently than classical computers. This makes it an interesting approach to deal with the computational challenges associated with PSR.

A key step in PSR is finding the solution to Poisson's equation

Improve: At the core of PSR is the solution of a Poisson equation, which can be discretised into a large sparse linear system. Quantum algorithms for solving linear systems, such as the HHL algorithm and its successors, have demonstrated theoretical advantages for certain classes of large sparse systems. This suggests that quantum computing could potentially be applied to the Poisson solve within PSR. Therefore, this project will investigate whether quantum algorithms can be used to improve the efficiency of the Poisson solving stage of PSR, particularly for large-scale, high-resolution surface reconstruction.

1.2 Aim

The aim of this project is to investigate the application of quantum computing to Poisson Surface Reconstruction, with a particular focus on the Poisson equation at the center of the reconstruction method.

Improve: The project will explore whether quantum algorithms for solving large sparse linear systems can be applied to this problem and evaluate their potential to improve the computational efficiency of reconstructing high-resolution surfaces from large point clouds.

2 Classical Implementation

2.1 Algorithm Overview

The proposed surface reconstruction method takes a set of oriented two-dimensional scan points as input and reconstructs a continuous surface from these discrete observations. Each scan point is associated with an estimated surface normal which is used to construct a vector field on a regular computational grid. The divergence of this vector field is then calculated and used as the right-hand side of a Poisson equation. The Poisson equation is solved in the Fourier domain using the Fast Fourier Transform (FFT), producing an implicit scalar function representing the reconstructed surface. Finally, an iso-value is estimated from the reconstructed function at the original scan locations, and the corresponding iso-contour is extracted to obtain the reconstructed surface coordinates.

The overall algorithm used can be summarised as:

Oriented point cloud $\rightarrow$ Vector field $\rightarrow$ Divergence $\rightarrow$ Poisson solution $\rightarrow$ Iso-value $\rightarrow$ Surface contour.

The method operates entirely in two dimensions. Consequently, the final reconstructed surface is represented by one or more two-dimensional contours rather than a three-dimensional triangular mesh.

2.2 Point Cloud and Computational Grid

2.3 Vector Field Construction

2.4 Divergence Approximation

2.5 Fourier Based Poisson Solver

2.6 Fourier Based Poisson Solver

2.7 Iso-value Estimation and Surface Reconstruction

2.8 Computational Complexity and Scalability

3 Sensitivity and Performance Analysis

3.1 Accuracy Evaluation

Intersection over Union

The Intersection over Union (IoU) is a metric used to measure how much two surfaces overlap.

$$\text{IoU} = \frac{\text{Intersection}}{\text{Union}} = \frac{|A \cap B|}{|A \cup B|}$$

where,

- Intersection is the area shared by the actual surface and the predicted surface.
- Union is the total area covered by either region.

Chamfer Distance

4 Quantum Poisson Solver